AI for Bharat Hackathon

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TEAM NAME

SupaDev

PROBLEM STATEMENT

Tool Fragmentation & Accessibility in Bharat

TEAM LEADER NAME

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Brief about the Idea

The Core Concept

SupaDev is an AI-powered Agentic IDE designed specifically for Bharat's developers. It operates offline-first, ensuring privacy and accessibility even with poor internet connectivity in Tier 2/3 cities.

How is it different?

Unlike standard autocomplete tools (e.g., Copilot), SupaDev features autonomous agents that Research, Architect, Code, and QA. It is a full IDE replacement, not just a plugin.

Solving the Problem

- **Eliminates Fragmentation:** Reduces context switching between 5-8 tools, saving ~40% productivity.
- **Bridge the Gap:** Provides mentorship to junior devs via AI agents.
- **Offline Capability:** Runs locally to bypass connectivity issues.

> **USP:** Privacy-First Architecture + 6-Stage Autonomous Agentic Pipeline.

List of features offered by the solution

AI-Native Core

Features Ghost completions, Smart Documentation (Option+Click), and a Natural Language Command Bar that converts "Make this blue" into CSS code instantly.

Agentic Pipelines

A 6-stage autonomous workflow where agents handle Research, Architecture, Coding, Verification, and QA before Human approval.

Full IDE Replacement

Not a plugin. A complete environment with built-in Terminal, Git integration, Debugger (LLDB), and support for 35+ languages via Tree-sitter.

Privacy-First

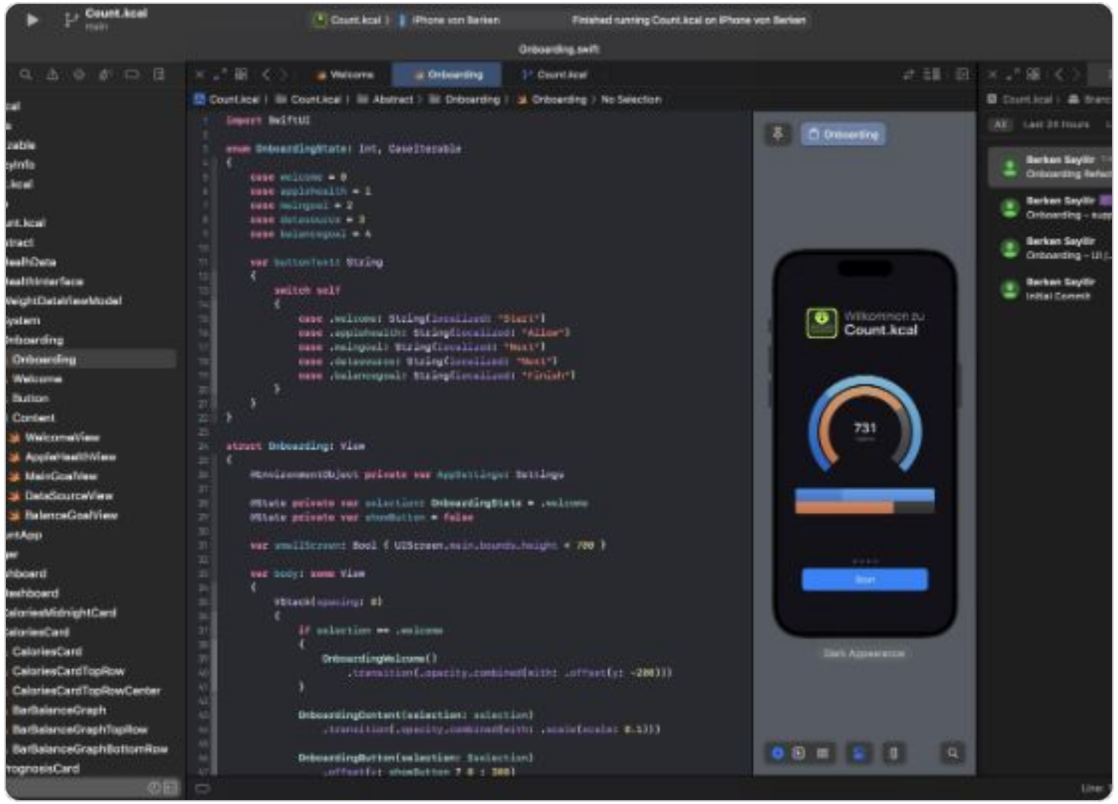
Code stays local. Intelligent router sends only necessary prompts to AWS Bedrock, ensuring IP protection and no telemetry.

Process Flow: Agentic Pipeline

The SupaDev workflow moves beyond simple code suggestion into a structured, autonomous pipeline:

Wireframes of the proposed solution

SupaDev Editor Interface



The interface is designed to be familiar to VS Code users but streamlined for agentic interactions. The "Ghost Text" (purple overlay) shows agent suggestions in real-time.

Integrated Terminal

Built-in terminal with AI command injection.

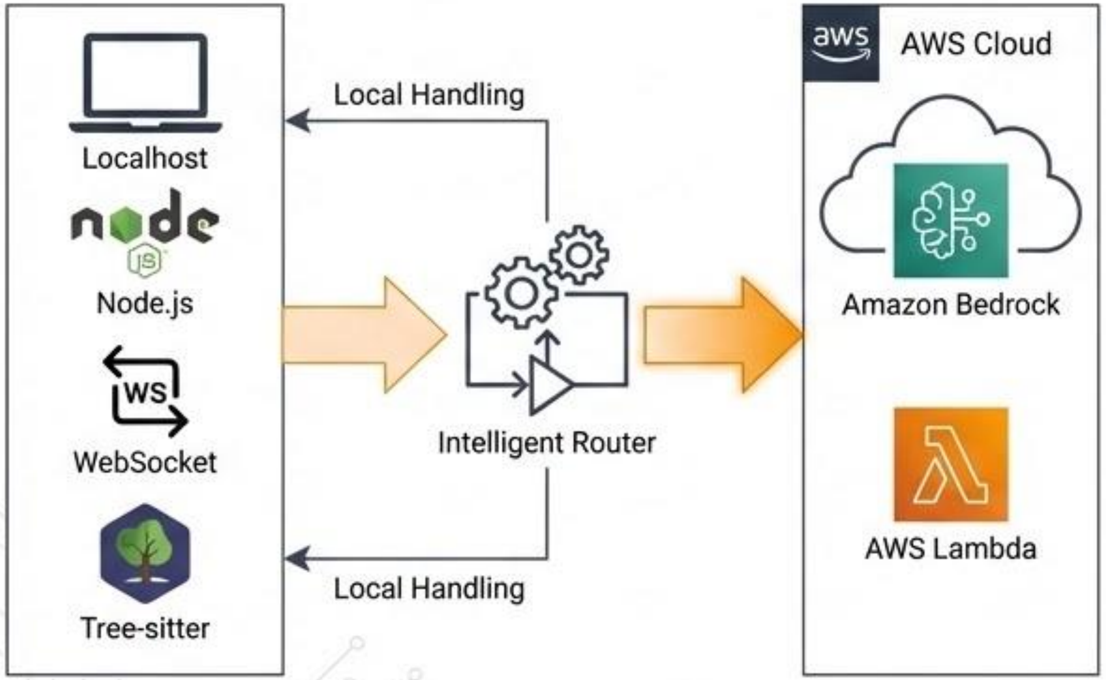
Git Graph

Visual commit history and hunk-level staging managed by agents.

Intelligent Debugger

One-click "Fix It" button that analyzes stack traces and applies patches.

Architecture diagram of the proposed solution



Privacy-First Hybrid Architecture

The architecture is split between local execution and cloud intelligence to ensure speed and privacy.

- **Localhost:** The IDE runs Node.js + WebSocket locally. Code analysis (Tree-sitter) happens on the device.
- **Intelligent Router:** Determines if a task can be handled locally or requires Cloud AI.
- **AWS Cloud:** Heavy lifting is offloaded to Amazon Bedrock (Claude/Llama models) and AWS Lambda for scalable agent execution.

Technologies to be used in the solution

Core Frameworks

- **Swift (AppKit/SwiftUI):** For native macOS performance.
- **Tree-sitter:** For high-performance syntax highlighting and code parsing.
- **LLDB:** Integrated debugging infrastructure.
- **Node.js:** Local backend and extension host.

AWS Cloud Stack

- Amazon Bedrock
- AWS Lambda
- Amazon S3
- AWS CodePipeline

Estimated Implementation Cost Breakdown

Breakdown of estimated costs for developing, deploying, and maintaining the SupaDev platform.

Development & Initial Setup

- AI Model Fine-tuning & Training: ₹1,50,000
- Backend & API Development: ₹2,00,000
- IDE Extension Development: ₹1,00,000
- Testing & QA: ₹50,000

Operational Costs (Monthly)

- AWS Bedrock (AI Inference): ₹15,000/mo
- AWS Lambda (Serverless Compute): ₹5,000/mo
- Amazon S3 (Storage): ₹1,000/mo
- Maintenance & Updates: ₹10,000/mo

Infrastructure & Tooling (One-time)

- DevOps & CI/CD Pipeline Setup: ₹75,000
- Security Audits & Compliance: ₹50,000
- Monitoring & Logging Setup: ₹25,000

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Thank You